

Maven

Maven is DevOps tool that help to build, Package, and manage software projects by automating the build process, managing dependencies, and providing a standardized project structure.

MAIN FILE OF MAVEN (pom.xml):

- The pom.xml (Project Object Model) is the core configuration file in a **M**aven project. It contains project metadata, dependencies, and build configuration.

PROJECT INFORMATION:

- The pom.xml file holds details about the project, including the group ID, artifact ID, version, and dependencies required for the build.

BUILD TOOL:

- **M**aven is a build automation tool primarily used for Java projects. It simplifies the build process by managing project dependencies and automating tasks such as compiling code, running tests, and packaging the project.

ARTIFACT CREATION:

- **M**aven is used to create deployable artifacts such as .jar (Java Archive), .war (Web Application Archive), and .ear (Enterprise Application Archive) files.

JAVA-BASED PROJECTS:

- **M**aven is widely used in the Java ecosystem to manage dependencies, streamline builds, and automate project management tasks.

RELEASE DATE:

- Maven was officially released on 13 July 2014.

WRITTEN IN JAVA:

- Maven is developed using Java, specifically written in Java version 1.8.0.

PRE-MAVEN JAVA BUILD PROCESS:

1. RAW: .JAVA (SOURCE CODE):

Java source code written by the programmer, each file representing a class or interface.

2. COMPILE:

The .java files are compiled into .class files (bytecode) using a Java compiler like javac.

3. CLASS FILES:

These compiled files can be executed by the JVM but are not yet packaged for distribution.

4. BUILD .JAR (JAVA ARCHIVE):

.class files are bundled into a .jar file, a package containing Java classes and resources for distribution.

5. PACKAGE INTO .WAR (WEB APPLICATION ARCHIVE):

For web applications, .class files and resources are packaged into a .war file, ready for deployment on a server.

ROLE OF MAVEN:

Maven automates this entire process, handling dependencies, compiling code, and packaging into .jar and .war files. It simplifies build and deployment, ensuring all libraries and resources are included automatically.

ARTIFACTS:

Definition: The final product of the build process, such as .jar, .war, or .ear files. The purpose will be that the files will be ready for deployment or distribution.

.EAR (ENTERPRISE ARCHIVE):

Definition: An .ear file packages multiple .jar and .war files together. Used for large enterprise applications that require both backend services and web components.

HOW INSTALL MAVEN:

1.CREATE EC2

2) INSTALLATION OF MAVEN:

FOR LINUX

```
yum install git java-1.8.0-openjdk maven tree -y
```

FOR DEBIAN

```
apt install git java-1.8.0-openjdk maven tree -y
```

MAVEN COMMANDS

TO COMPILE THE SOURCE CODE:

⇒ **mvn compile**

TO TEST THE CODE:

⇒ **mvn test**

TO CREATE ARTIFACT (project directory)

⇒ **mvn package**

TO CREATE ARTIFACT (maven home dir .m2)

⇒ **mvn install**

TO PERFORM ENTIRE GOALS

⇒ **mvn clean package**

TO DELETE TARGET:

⇒ **mvn clean**

MAVEN	ANT
Convention over configuration	Configuration over convention
Follows a defined build lifecycle with predefined phases (e.g., compile, test, package)	No predefined lifecycle; requires manual task definition
Built-in dependency management through a central repository (via pom.xml)	No built-in dependency management; relies on third-party tools (like Apache Ivy)
XML-based pom.xml file where configurations are declarative and standardized	XML-based build.xml file where configurations are procedural and manual
Uses plugins to handle specific tasks automatically	Uses tasks to define build steps, requiring more manual setup
Enforces a standard directory structure (e.g., src/main/java)	No enforced directory structure; developers must define it manually
Uses central repositories (e.g., Maven Central) to manage and download dependencies	No default repository support; dependencies must be managed manually or with external tools
Strong integration with modern IDEs (e.g., Eclipse, IntelliJ IDEA)	Also integrates well, but requires more manual configuration

WHAT IF BUILD FAILS?

1. CHECK JAVA VERSION FOR MAVEN
2. CHECK POM.XML
3. CHECK THE CODE

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